

Mixed-Initiative Interaction with Computational Generative Systems

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ABSTRACT

Machine learning models provide functions to transform and generate image and text data. This promises powerful applications but it remains unclear how users can interact with these models. With my research, I focus on designing, implementing, and evaluating functional prototypes for understanding human-AI interactions. Methodologically, I focus on web-based experiments with a mixed-methods approach. Furthermore, I use these prototypes and generative models as a material to understand fundamental concepts in human-AI interactions, such as initiative, intent, and control. In an already conducted study, for example, I showed that the levels of initiative and control afforded by the UI influence perceived authorship when writing text. For the future, I plan to carry out more studies on collaborative writing. With my dissertation, I contribute to how we will build human-AI interactions and how we will collaborate with computational generative systems in future.

CCS CONCEPTS

• **Human-centered computing** → **Text input; Empirical studies in HCI**; • **Computing methodologies** → **Natural language processing**.

KEYWORDS

human-ai interaction, mixed-initiative, generative systems, writing, typing, language model, text generation, control, initiative, intent, functional prototypes

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1 INTRODUCTION

Latest developments in the domain of machine learning contributed models that can transform data, or generate new data. Various of these models have been published (e.g. [4, 19, 20, 23, 30]). Popular models in this regard are, for example, DALL-E [21] and GPT-3 [4]. These models are able to generate image, respectively text data.

The GPT model can predict text based on input text. The output of the model is an extended version of the input text. More broadly, these machine learning models are considered artificial intelligence (AI). When such models are released, the knowledge about their capabilities and the use-cases they could be applied to are rather vague. Since machine learning models are technical contributions, papers about them report naturally about the basic functionality and evaluate the performance. However, it remains unclear how we can integrate such models into existing workflows, if there exist more use-cases and functionality, and how people could effectively interact with them.

HCI researchers work to answer these aspects from different points of view. Work by Yang et al. [33, 34], for example, explores the challenges designing for Human-AI interactions, and NLP applications specifically [32]. As a practical outcome, Amershi et al. [1] have published human-AI guidelines as a resource for designers that work on human-AI applications. When involving generative models in human-AI interactions, it becomes plausible to attribute a more active role to the system. In collaborative writing, for instance, the generative system could become the author and actively collaborate on a text. Such human-AI applications are known as mixed-initiative interactions [11]. In mixed-initiative interactions, initiative switches between the user and the system.

Designing and building human-AI applications with mixed-initiative interactions requires us to understand fundamental concepts like intent, initiative, and control. In particular, we might think about these concepts as parameters that can be adjusted for human-AI interaction. This might influence the usage of applications with intelligent features. With my work, I focus on investigating how computational generative systems, in particular generative text models, can be integrated into workflows for writers. Furthermore, I explore new types of interaction for collaborative writing with AI that are enabled by these intelligent systems. To conduct my research, I design, implement, and evaluate functional prototypes in web-based experiments. Moreover, I use these human-AI interaction prototypes as a material to understand fundamental concepts in interactions, such as initiative and control. With the results of my research I provide a basis to inform future human-AI applications and to shape how we will collaborate with intelligent systems.

2 RELATED WORK

Work related to my dissertation is located at the intersection of HCI and AI. It includes technical contributions on machine learning models and design-related HCI contributions. Furthermore, a strand in HCI research focuses explicitly on human-AI interaction with a scope on interacting with text models. In the following, I highlight

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some of the works within these areas and how they link to my work.

2.1 Technical Contributions

With advances in deep learning, research on machine learning contributes work on generative models. In particular, these models are able to extend text, based on a given input. Techniques to process text are known as natural language processing (NLP). Depending on the input text, these models cover various NLP tasks, such as translating, question answering, and paraphrasing. Basic research to enable these generative models were Transformer networks by Vaswani et al. [30]. Different to existing encoder-decoder architectures in the past, recurrent layers were replaced by self-attention mechanisms. Based on this Transformer networks is the language model GPT-3 [4]. This model is the successor of GPT-2 [19]. Other Transformer models are Megatron [26] and OPT [36]. These transformer networks are all considered state-of-the-art for extending text. With zero-shot and few-shot learning [4] approaches, their capability to extend text can be steered to fulfill a certain tasks, for example, translating text from one language into another. This technique is known as prompting [22]. Methods such as prompting span the gap between the rather technical domain on machine learning and HCI, the latter focuses on approaches to make these models usable from an user perspective.

2.2 HCI and Human-AI Contributions

Besides recent efforts on making prompting explorable [28, 31], HCI researchers are investigating human-AI interaction from various perspectives. From a designer's perspective, work by Dove et al. [7] identified challenges of using machine learning models as a material for UX design. Further work in this direction was carried out by Yang et al. [33, 34]. In more specific work on language based models, Yang et al. [32] looked into sketching methods for NLP enabled tools and features. These papers see models primarily as a design material and analyze challenges for designers building intelligent tools. Core challenges are, for example, uncertainty and thinking into the abilities of models. Intelligent tools and user interfaces oftentimes provide mixed-initiative interactions [11]. In such interactions, not only the human has initiative to carry out a task, but also the system. To take into consideration the fundamental concept of initiative when building intelligent tools, Horvitz [11] introduced 12 principles to integrate intelligent features effectively. Similar to that, work by Amershi et al. [1] proposed 18 design-guidelines for human-AI interaction. These human-AI interaction guidelines provide examples for applications and contexts to which they should be applied. Design practitioners can potentially include these guidelines into their design workflow.

Going beyond the design stage, another strand on human-AI interaction designs, implements, and evaluates functional prototypes. As highlighted with the following examples, this work is dedicated to interaction with (generative) language models. Collaborative writing with language models in latest works oftentimes happens through suggestions (e.g. [2, 3, 5, 15]). Research on such text suggestions has shown that a sentiment bias of suggestions can lead to biased writing overall [2], writing short descriptions with suggestions lead to shorter text and less broad use of vocabulary

[3]. Work by Buschek et al. [5] on writing emails with multiple suggested phrases revealed that users benefited finding ideas but became slower. Furthermore, they showed that non-native speakers benefited from more suggestions displayed. As these works show, writing with AI can influence aspects such as inspiration and efficiency. Other work investigate collaborative writing with AI in specific contexts, such as creative and science writing. Work by Gero et al. [8], investigated how language models can support science writing. Writing with a generative model within their prototype led writers to craft detailed sentences but also to gain an understanding of how to engage readers and of the readers' perspective overall. Other work by Singh et al. [27] on inspiring writers showed that offering system generated multimedia stimuli by analyzing the written text helped to inspire the writer, even without accepting system generated suggestions. Furthermore, Lee et al. [13] published a dataset on creative writing for revealing generative capabilities of GPT-3 [4]. In a study by Yuan et al. [35] explored how people might use generative models in the process of creative writing. They showed that controls seem to improve helpfulness and enjoyableness.

2.3 Influence on my Research

The technical contributions as well as the HCI contributions inform, inspire, or share similarities to my work. I chose to conduct experiments mostly with functional prototypes to go beyond the design stage. This approach enables me to explore novel interactions in a realistic context and to understand fundamental concepts. In a recent study, for example, I contrasted writing with AI generated suggestions and automatic writing of AI. In this study we found effects on text length, wording, and perceived authorship [15]. Furthermore, in an ongoing project I work on a software framework with the goal to make research on human-AI interaction more convenient and transparent. Moreover, I look into different types of intelligent UI modalities in another project. In Section 5 I will describe the findings of my work so far in more detail.

3 RESEARCH GOALS

The main research goals of my work are as follows:

- Explore novel interactions with computational generative systems.
- Explore possibilities to integrate these systems into workflows for writers.
- Understand fundamental concepts in interaction, such as initiative and control.

4 METHODS

4.1 Iterative Design & Functional Prototypes

Central to my research is building functional prototypes for understanding the interactions with computational generative systems, with a focus on collaborative writing with AI. This approach allows me to investigate realistic interactions.

For designing these prototypes and interactions, I follow an iterative design approach. I inform this process by brainstorming sessions, formative surveys, expert groups, and interviews. After some cycles of sketching and understanding the main interaction

Aspect	Description
User Interface	The interface holds a field for input. Generated objects are placed near the input.
Workflow	One or more suggestions are generated interactively and are part of the workflow
User Decision	The user can decide to accept a suggestion or not.
Editing	The user can further edit the suggestion object.
Information	Partial user input serves as input for the system. The AI's prediction is conditioned on the input to extend it.

Table 1: Table taken from [14]. With this work, we identified five aspects of autocompletion in interactive applications. Furthermore, we propose that these aspects might be transferred to other human-AI interactions as long as their goal is to extend partial user input.

of a prototype, I switch rather quickly to implementing. Trying out interactions with a realistic user interface enables us in the research group to discuss meaningful improvements to the prototype designs.

4.2 Mixed-Methods Approach

For evaluating these prototypes and exploring fundamental interaction concepts, I combine qualitative and quantitative methods. With this mixed-method approach, I can gather a more complete picture of the users interacting with the system. The interaction logs reveal the actual use of the prototype while questionnaires and interviews complete this picture by rather subjective feedback.

4.3 Web-Based Field Experiments

Most of my studies, I conduct as web-based field experiments. Field experiments are a combination of field studies and lab studies. They allow researchers to control some variables while having a more realistic context of usage. With a software framework, I implement the experimental designs, e.g. to make sure procedures are executed correctly by participants. Conducting web-based field experiments also allows me to scale studies on demand and reach broader samples of participants.

5 RESULTS

At the time of writing this paper, I have published two full papers in the context of my dissertation. With these papers, I investigated various aspects of human-AI interaction. My contributions cover a conceptual lens on interacting with generative AI and analyzing interaction methods. Furthermore, I have developed a software system as a methodological basis for conducting studies with functional prototypes (not yet published). In the following, I describe these three works in more detail.

5.1 Examining Autocompletion as a Basic Concept for Interaction with Generative AI

Through a conceptual lens, we dissected the user interface and interaction elements of textual autocompletion and other intelligent applications [14]. These intelligent applications were code completion, GUI sketching, and layouting. We compared elements across applications and found recurring patterns. For this analysis, we took a user-centered perspective and identified five aspects which can support designers to pragmatically approach the interaction design for generative AI. Two of these aspects are workflow and

editing. The aspect of workflow highlights a continuous interaction between the system and the user. The aspect of editing highlights that the user is able to edit a system generated suggestion until it fits the intent. See Table 1 for a complete overview of the five aspects. Based on these finding, we argue that an inherent role of intelligent systems is to extend and complete user input. With this conceptual approach we hope to support designers in informing and creating interactions for generative AI.

5.2 Suggestion Lists vs. Continuous Generation: Interaction Design for Writing with Generative Models

We designed and compared two interactive methods to write text with AI on mobile devices [15]. With these methods we manipulated the levels of initiative and control. Participants in this study had to write text with AI support in two methods. With the first method, people had to write text with AI continuously generating text. Here we emulated automatic writing word-by-word. With the second method, participants had to compose text from AI suggested phrases. The user interfaces of these methods can be seen in Figure 1. We conducted a mixed-methods study and collected data on interactions, ratings, and interviews. Overall, we observed these interactions to affect the text length and wording of written text as well as perceived authorship. In particular, when composing text from AI suggestions, people had the least keyboard interactions but still perceived themselves as authors. We discussed and reflected on these results, for example, in light of role frameworks for co-creativity [9, 12, 17] and propose to integrate initiative and control as explicit parameters into such frameworks. With these findings, we highlight the importance of UI design decisions for tools implementing human-AI co-creativity and how these decisions impact user experience and output.

5.3 HCI Experiment Framework

Prototypes in HCI research can be build as web applications. In particular, web applications have been used for prototyping in recent research on human-AI interaction (e.g. [2, 6, 8, 13, 15, 24, 29]). A benefit of such web prototypes is that they can be evaluated in online studies, for example, field experiments [25]. Setting up proper experiment procedures, combining the prototypes with questionnaires, and integrating data logging can be tedious and complex. To reduce required effort setting up web-based field experiments, we developed a software framework. In contrast to existing toolkit and

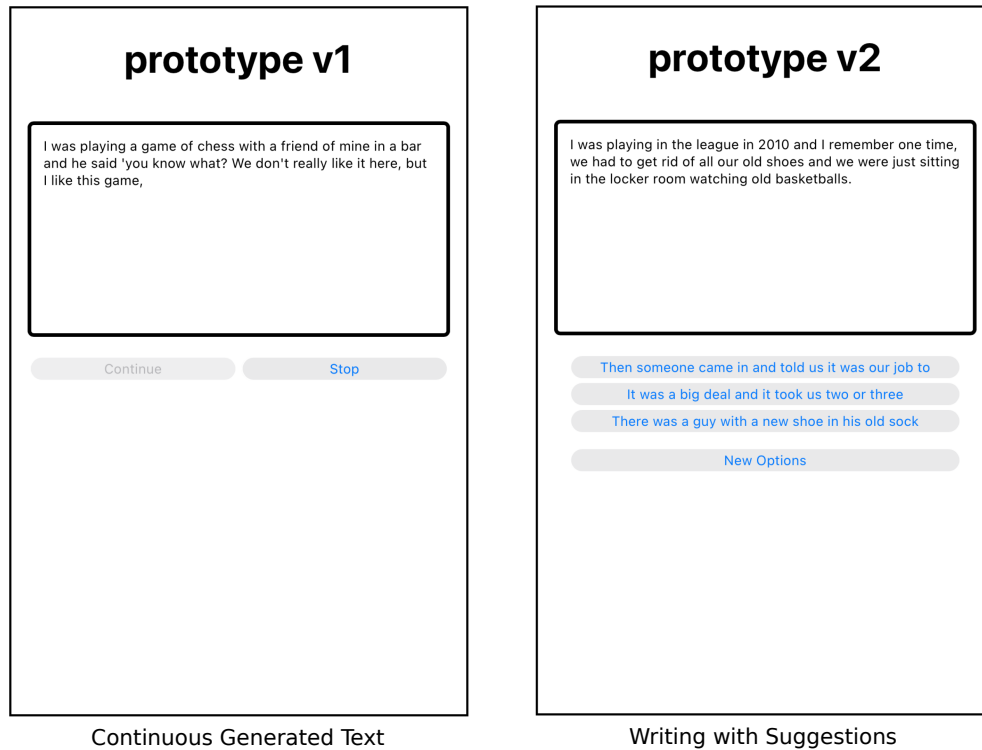


Figure 1: Figure taken from [15]. These screenshots show the final versions of the prototypes we implemented as mobile web-applications. The interface on the left integrates AI writing continuously word-by-word. The interface on the right allows the user to compose text from AI generated phrases.

frameworks (e.g. [10, 16, 18]), our framework has a focus on HCI studies including prototypes with functional interaction. It allows HCI researchers to setup, duplicate, replicate, and share their studies as well as logging data. All of these features are particularly useful for research that builds on multiple subsequent studies, collecting qualitative and quantitative data. With this contribution I provide systematic methods for web-based HCI research. Furthermore, the system aids transparency since study designs can be shared in a formal way.

6 FUTURE WORK & CHALLENGES

To complement my work, I plan for the near future to conduct further studies on writing with AI, for example, looking into novel interaction methods to actively support writers in creating texts. Since my prior work has highlighted the importance of human control I will focus on strategies to compose text to minimize active writing. My vision in this regard is to provide a structured UI for writers to create text hand-in-hand with AI. At the same time I plan to critically reflect on the roles of writers and their intents. Beyond that, I plan to incorporate other topics to my research agenda. One possibility could be to adapt my knowledge from the text domain to the image domain. Another direction would be to create a general theoretical framework of human-AI interaction.

A major challenge in future work will be to carry out long-term studies. In recent work, I build and evaluated the prototypes for specific use cases and tasks and did not run long-term studies. Such long-term studies would enable us to derive more general findings and to further improve human-AI interaction extending human skills.

7 RESEARCH SITUATION

I started with my doctorate in spring 2020 under the supervision of Daniel Buschek at the University of Bayreuth. My PhD position is funded as part of a research group with a focus on research at the intersection of HCI and AI. I do an individual doctorate without taking part in a structured program. This is the traditional setup in Germany. Currently, I am in my third year and plan to graduate after four years in 2024. After that, I primarily aim to stay in academia, but could also imagine to work as an industry researcher.

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REFERENCES

- [1] Saleema Amershi, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, Eric Horvitz, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, and Paul N. Bennett. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems - CHI '19*. ACM Press, New York, NY, USA, 1–13. <https://doi.org/10.1145/3290605.3300233>
- [2] Kenneth C. Arnold, Krysta Chauncey, and Krzysztof Z. Gajos. 2018. Sentiment Bias in Predictive Text Recommendations Results in Biased Writing. In *Proceedings of the 44th Graphics Interface Conference (GI '18)*. Canadian Human-Computer Communications Society, Toronto, Canada, 42–49. <https://doi.org/10.20380/GI2018.07>
- [3] Kenneth C. Arnold, Krysta Chauncey, and Krzysztof Z. Gajos. 2020. Predictive text encourages predictable writing. In *Proceedings of the 25th International Conference on Intelligent User Interfaces (IUI '20)*. Association for Computing Machinery, Cagliari, Italy, 128–138. <https://doi.org/10.1145/3377325.3377523>
- [4] Tom B. Brown, Benjamin Mann, Nick Ryder, Melanie Subbiah, Jared Kaplan, Prafulla Dhariwal, Arvind Neelakantan, Pranav Shyam, Girish Sastry, Amanda Askell, Sandhini Agarwal, Ariel Herbert-Voss, Gretchen Krueger, Tom Henighan, Rewon Child, Aditya Ramesh, Daniel M. Ziegler, Jeffrey Wu, Clemens Winter, Christopher Hesse, Mark Chen, Eric Sigler, Mateusz Litwin, Scott Gray, Benjamin Chess, Jack Clark, Christopher Berner, Sam McCandlish, Alec Radford, Ilya Sutskever, and Dario Amodei. 2020. Language Models are Few-Shot Learners. *arXiv:2005.14165 [cs]* (July 2020). <http://arxiv.org/abs/2005.14165> arXiv: 2005.14165.
- [5] Daniel Buschek, Martin Zürn, and Malin Eiband. 2021. The Impact of Multiple Parallel Phrase Suggestions on Email Input and Composition Behaviour of Native and Non-Native English Writers. *arXiv:2101.09157 [cs]* (Jan. 2021). <https://doi.org/10.1145/3411764.3445372> arXiv: 2101.09157.
- [6] John Joon Young Chung, Woosook Kim, Kang Min Yoo, Hwaran Lee, Eytan Adar, and Minsuk Chang. 2022. TaleBrush: Sketching Stories with Generative Pretrained Language Models. In *CHI Conference on Human Factors in Computing Systems*. ACM, New Orleans LA USA, 1–19. <https://doi.org/10.1145/3491102.3501819>
- [7] Graham Dove, Kim Halskov, Jodi Forlizzi, and John Zimmerman. 2017. UX Design Innovation: Challenges for Working with Machine Learning as a Design Material. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. ACM, New York, NY, USA, 278–288. <https://doi.org/10.1145/3025453.3025739>
- [8] Katy Ilonka Gero, Vivian Liu, and Lydia Chilton. 2022. Sparks: Inspiration for Science Writing using Language Models. In *Designing Interactive Systems Conference*. ACM, Virtual Event Australia, 1002–1019. <https://doi.org/10.1145/3532106.3533533>
- [9] Matthew Guzdial and Mark Riedl. 2019. An Interaction Framework for Studying Co-Creative AI. *arXiv:1903.09709 [cs]* (March 2019). <http://arxiv.org/abs/1903.09709> arXiv: 1903.09709.
- [10] Felix Henninger, Yury Shevchenko, Ulf Kai Mertens, Pascal J. Kieslich, and Benjamin E. Hilbig. 2019. *lab.js: A free, open, online study builder*. preprint. PsyArXiv. <https://doi.org/10.31234/osf.io/fqr49>
- [11] Eric Horvitz. 1999. Principles of mixed-initiative user interfaces. In *Proceedings of the SIGCHI conference on Human Factors in Computing Systems (CHI '99)*. ACM Press, New York, NY, USA, 159–166. <https://doi.org/10.1145/302979.303030>
- [12] Anna Kantosalo and Hannu Toivonen. 2016. Modes for Creative Human-Computer Collaboration: Alternating and Task-Divided Co-Creativity. In *Proceedings of the Seventh International Conference on Computational Creativity (ICCC 2016)*. Sony CSL, Paris, 77–84. <http://www.computationalcreativity.net/iccc2016/wp-content/uploads/2016/01/Modes-for-Creative-Human-Computer-Collaboration.pdf>
- [13] Mina Lee, Percy Liang, and Qian Yang. 2022. CoAuthor: Designing a Human-AI Collaborative Writing Dataset for Exploring Language Model Capabilities. In *CHI Conference on Human Factors in Computing Systems*. ACM, New Orleans LA USA, 1–19. <https://doi.org/10.1145/3491102.3502030>
- [14] Florian Lehmann and Daniel Buschek. 2021. Examining Autocompletion as a Basic Concept for Interaction with Generative AI. *i-com* 19, 3 (Jan. 2021), 251–264. <https://doi.org/10.1515/icom-2020-0025>
- [15] Florian Lehmann, Niklas Markert, Hai Dang, and Daniel Buschek. 2022. Suggestion Lists vs. Continuous Generation: Interaction Design for Writing with Generative Models on Mobile Devices Affect Text Length, Wording and Perceived Authorship. In *Mensch und Computer 2022*. ACM, Darmstadt Germany, 192–208. <https://doi.org/10.1145/3543758.3543947>
- [16] Sebastiaan Mathôt, Daniel Schreij, and Jan Theeuwes. 2012. OpenSesame: An open-source, graphical experiment builder for the social sciences. *Behavior Research Methods* 44, 2 (June 2012), 314–324. <https://doi.org/10.3758/s13428-011-0168-7>
- [17] Santiago Negrete-Yankelevich and Nora Morales-Zaragoza. 2014. The apprentice framework: planning and assessing creativity. In *Proceedings of the Seventh International Conference on Computational Creativity (ICCC 2014)*. computationalcreativity.net, Ljubljana, 280–283. http://computationalcreativity.net/iccc2014/wp-content/uploads/2014/06/13.4_NegreteYankelevich.pdf
- [18] Jonathan W. Peirce. 2007. PsychoPy—Psychophysics software in Python. *Journal of Neuroscience Methods* 162, 1–2 (May 2007), 8–13. <https://doi.org/10.1016/j.jneumeth.2006.11.017>
- [19] Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. 2019. Language Models are Unsupervised Multitask Learners. (2019), 24. https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf
- [20] Colin Raffel, Noam Shazeer, Adam Roberts, Katherine Lee, Sharan Narang, Michael Matena, Yanqi Zhou, Wei Li, and Peter J. Liu. 2020. Exploring the Limits of Transfer Learning with a Unified Text-to-Text Transformer. *Journal of Machine Learning Research* 21, 140 (2020), 1–67. <http://jmlr.org/papers/v21/20-074.html>
- [21] Aditya Ramesh, Mikhail Pavlov, Gabriel Goh, Scott Gray, Chelsea Voss, Alec Radford, Mark Chen, and Ilya Sutskever. 2021. Zero-Shot Text-to-Image Generation. In *Proceedings of the 38th International Conference on Machine Learning (Proceedings of Machine Learning Research, Vol. 139)*, Marina Meila and Tong Zhang (Eds.). PMLR, 8821–8831. <https://proceedings.mlr.press/v139/ramesh21a.html>
- [22] Laria Reynolds and Kyle McDonell. 2021. Prompt Programming for Large Language Models: Beyond the Few-Shot Paradigm. In *Extended Abstracts of the 2021 CHI Conference on Human Factors in Computing Systems*. ACM, Yokohama Japan, 1–7. <https://doi.org/10.1145/3411763.3451760>
- [23] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. 2022. High-Resolution Image Synthesis With Latent Diffusion Models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 10684–10695.
- [24] Oliver Schmitt and Daniel Buschek. 2021. CharacterChat: Supporting the Creation of Fictional Characters through Conversation and Progressive Manifestation with a Chatbot. In *Creativity and Cognition*. ACM, Virtual Event Italy, 1–10. <https://doi.org/10.1145/3450741.3465253>
- [25] William R. Shadish, Thomas D. Cook, and Donald T. Campbell. 2002. *Experimental and Quasi-Experimental Designs for Generalized Causal Inference*. Houghton, Mifflin and Company, Boston, MA, US. Pages: xxi, 623.
- [26] Mohammad Shoeybi, Mostafa Patwary, Raul Puri, Patrick LeGresley, Jared Casper, and Bryan Catanzaro. 2020. Megatron-LM: Training Multi-Billion Parameter Language Models Using Model Parallelism. <http://arxiv.org/abs/1909.08053> arXiv:1909.08053 [cs].
- [27] Nikhil Singh, Guillermo Bernal, Daria Savchenko, and Elena L. Glassman. 2022. Where to Hide a Stolen Elephant: Leaps in Creative Writing with Multimodal Machine Intelligence. *ACM Transactions on Computer-Human Interaction* (Feb. 2022), 3511599. <https://doi.org/10.1145/3511599>
- [28] Hendrik Strobelt, Albert Webson, Victor Sanh, Benjamin Hoover, Johanna Beyer, Hanspeter Pfister, and Alexander M. Rush. 2022. Interactive and Visual Prompt Engineering for Ad-hoc Task Adaptation With Large Language Models. *IEEE Transactions on Visualization and Computer Graphics* (2022), 1–11. <https://doi.org/10.1109/TVCG.2022.3209479>
- [29] Kashyap Todi, Luis A. Leiva, Daniel Buschek, Pin Tian, and Antti Oulasvirta. 2021. Conversations with GUIs. In *Designing Interactive Systems Conference 2021*. ACM, Virtual Event USA, 1447–1457. <https://doi.org/10.1145/3461778.3462124>
- [30] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Proceedings of the 31st International Conference on Neural Information Processing Systems (NIPS'17)*. Curran Associates Inc., Long Beach, California, USA, 6000–6010.
- [31] Tongshuang Wu, Michael Terry, and Carrie J. Cai. 2021. AI Chains: Transparent and Controllable Human-AI Interaction by Chaining Large Language Model Prompts. *arXiv:2110.01691 [cs]* (Oct. 2021). <http://arxiv.org/abs/2110.01691> arXiv: 2110.01691.
- [32] Qian Yang, Justin Cranshaw, Saleema Amershi, Shamsi T. Iqbal, and Jaime Teevan. 2019. Sketching NLP: A Case Study of Exploring the Right Things To Design with Language Intelligence. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems - CHI '19*. ACM Press, New York, NY, USA, 1–12. <https://doi.org/10.1145/3290605.3300415>
- [33] Qian Yang, Alex Scuito, John Zimmerman, Jodi Forlizzi, and Aaron Steinfeld. 2018. Investigating How Experienced UX Designers Effectively Work with Machine Learning. In *Proceedings of the 2018 Designing Interactive Systems Conference (DIS '18)*. Association for Computing Machinery, Hong Kong, China, 585–596. <https://doi.org/10.1145/3196709.3196730>
- [34] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining Whether, Why, and How Human-AI Interaction Is Uniquely Difficult to Design. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)*. ACM Press, New York, NY, USA, 1–13. <https://doi.org/10.1145/3313831.3376301>
- [35] Ann Yuan, Andy Coenen, Emily Reif, and Daphne Ippolito. 2022. Wordcraft: Story Writing With Large Language Models. In *27th International Conference on Intelligent User Interfaces*. ACM, Helsinki Finland, 841–852. <https://doi.org/10.1145/3490099.3511105>
- [36] Susan Zhang, Stephen Roller, Naman Goyal, Mikel Artetxe, Moya Chen, Shuohui Chen, Christopher Dewan, Mona Diab, Xian Li, Xi Victoria Lin, Todor Mihaylov,

Myle Ott, Sam Shleifer, Kurt Shuster, Daniel Simig, Punit Singh Koura, Anjali Sridhar, Tianlu Wang, and Luke Zettlemoyer. 2022. OPT: Open Pre-trained

Transformer Language Models. <http://arxiv.org/abs/2205.01068> arXiv:2205.01068 [cs].